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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT PRESENTATION

CSA1614 - Data Warehousing and Data Mining

Intelligent Insurance Claim Analytics and Risk Assessment Using Anomaly Detection and Predictive Claims Classification

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Abstract

- Insurance companies process large volumes of claims, making manual verification time-consuming and error-prone.
- Data warehousing integrates and stores insurance claim data in a structured format for analysis.
- Data preprocessing improves data quality by handling missing values, outliers, and inconsistent data.
- Anomaly detection techniques identify unusual and potentially fraudulent claim patterns.
- Machine learning algorithms classify insurance claims and predict their risk levels.
- The system aims to reduce manual effort, improve risk assessment, and support faster claim processing.

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INTRODUCTION

- Insurance claim analysis is essential for identifying suspicious claims and assessing risk.
- Manual claim verification can be slow and may overlook unusual claim patterns.
- Data mining techniques help discover hidden patterns and relationships in insurance claim data.
- Anomaly detection identifies claims that differ significantly from normal patterns.
- Predictive classification algorithms categorize claims into low-, medium-, and high-risk levels.
- The proposed system supports efficient claim analysis and data-driven decision-making.

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Module Overview

- Module 1: Insurance Data Preprocessing & Exploratory Analytics
- Module 2: Fraud Detection Using Anomaly Detection
- Module 3: Predictive Claims Classification & Risk Assessment

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